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Dynamic correlations and portfolio optimization in socially responsible investments: evidence from Indonesia and South Korea

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This study contributes to the literature on socially responsible investing by examining the diversification potential of green bonds and futures for a commodities' index, gold and treasury bonds, alongside the SRI-KEHATI Index and the Dow Jones Sustainability Korea Index (DJSKI) from 2014 to 2024. Asymmetric Dynamic Conditional Correlation (A-DCC) models with GJR-GARCH in conditional variance and out-of-sample analysis are employed to explore dynamic conditional correlations, optimal weights and hedge ratios for stocks portfolio that include the SRI-KEHATI Index or the DJSKI. Constructing portfolios with optimal weights proves to be more effective in risk reduction compared to the hedge ratio strategy. The portfolio with optimal weights containing commodities exhibited the highest risk-adjusted return for both countries during the pre-war COVID-19 pandemic period, while in the subsequent period, portfolios containing gold performed best. Our findings provide valuable insights for investors, portfolio and risk managers, policymakers and index managers, aiding in improved risk management and decision-making for optimal portfolio allocation.

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Introduction

Socially responsible investment (SRI) has become a cornerstone of global finance, addressing critical challenges such as climate change, sustainable development, and corporate governance reform. By integrating environmental, social, and governance (ESG) criteria into investment decisions, SRI aligns financial performance with ethical imperatives, promoting resilience in the face of global uncertainties¹. This alignment has gained urgency during recent crises, such as the COVID-19 pandemic and the Russia-Ukraine war, which have disrupted financial systems worldwide and underscored vulnerabilities in traditional financial markets. Investors increasingly prioritize corporate social responsibility (CSR) and non-financial impacts, such as carbon reduction and ethical business practices, highlighting a shift toward integrating social and environmental concerns into portfolio decisions (Bilbao-Terol et al., 2016; Junkus and Berry, 2015).

South Korea and Indonesia, as key economies within the ASEAN + 6 framework, exemplify the complex interplay between growth and sustainability in contrasting economic contexts. South Korea, as a developed economy with advanced ESG policies and robust financial markets, demonstrates the potential of integrating sustainability into financial practices. The Dow Jones Sustainability Korea Index (DJSKI) reflects South Korea's progress in aligning corporate and financial practices with global sustainability goals. Conversely, Indonesia, Southeast Asia's largest economy, faces challenges in implementing ESG principles within an emerging market context. The SRI-KEHATI index highlights Indonesia's growing commitment to addressing issues such as deforestation, renewable energy adoption, and social equity. These two economies provide an ideal framework for analyzing how differences in market maturity and regulatory environments shape ESG performance and its role in mitigating financial risks.

The ASEAN Economic Community (AEC) aimed to improve capital flows and financial integration across the region. However, empirical evidence suggests that ASEAN markets remain only partially integrated, offering potential diversification benefits for investors (Robiyanto et al., 2023b). Emerging markets such as Indonesia are crucial to the global economy due to their large market potential and significant commodity consumption, yet they present unique challenges for ESG integration (Dai, 2021).

Dong et al. (2021) highlight the vulnerability of small open economies like those in ASEAN + 6 to external shocks, while Chancharat and Sinlapates (2023) underscore the underexplored nature of ESG investments in these markets compared to regions such as the US, Europe, or BRICS. Combining SRI indices with commodities, green bonds and traditional financial instruments presents significant diversification opportunities, although the literature remains limited and has focused on developed markets (Shahid et al. 2023).

Several key questions remain underexplored in the literature: How do differences in economic maturity and regulatory frameworks affect the stability, performance, and risk mitigation capabilities of SRI portfolios in emerging and developed markets? To what extent are there assets that can act as effective diversification and hedging tools for SRI investment in periods of economic turbulence, such as the COVID-19 pandemic and the Russia-Ukraine war? What is the optimal asset allocation strategy that combines green bonds, commodities, gold, or treasury bonds with the SRI-KEHATI Index and the DJSKI to maximize diversification benefits and risk-adjusted returns?

Thus, the objective of this study is to analyze SRI in Indonesia and South Korea by examining the diversification potential of green bonds, commodities, gold, and treasury bonds, alongside the SRI-KEHATI Index and the DJSKI. This objective is achieved

through the application of an Asymmetric Dynamic Conditional Correlation (A-DCC) model with GJR-GARCH in conditional variance and an out-of-sample analysis. The latter enables daily predictions of portfolio weights and hedge ratios, a method that has not been previously applied in the SRI segment of these countries. To the best of our knowledge, no prior studies have analyzed each two SRI indices in combination with such a wide range of instruments using optimization and hedging strategies.

This paper adopts the perspective of an institutional investor considering passive investment strategies in two assets portfolios and builds upon the current literature in several ways to address existing gaps. Firstly, it provides updated information for the optimization of sustainable investment portfolios by analyzing new opportunities for risk-adjusted returns while including the COVID-19 crisis period and the ongoing Russia-Ukraine war, incorporating a wide range of potentially diversifying assets. Secondly, it is the first study on SRI portfolio construction in South Korea using the proposed methodology. It also expands previous research on Indonesia by comparing the risk reduction of two types of portfolios and incorporating new assets into these portfolios alongside the sustainable index. Thirdly, this study is the first to incorporate an out-of-sample, one-step-ahead forecasting analysis using a rolling A-DCC model with a fixed window in these countries. Finally, the evaluation extends to market indices, enabling a comparison of risk-adjusted performance between sustainable and conventional portfolios within and across both countries.

The analysis confirms that incorporating sustainable indices into portfolios, alongside assets such as gold, green bonds and commodities, offers significant diversification benefits. Our results show that all assets used contribute to reducing the risk of the individual SRI portfolio when used in optimally weighted portfolios, and that these are more effective in reducing risk compared to the hedge ratio strategy. The portfolio with optimal weights containing commodities exhibited the highest risk-adjusted return in both countries during the pre-war COVID-19 pandemic period, while in the subsequent period, portfolios containing gold performed best. Our findings emphasize the context-dependent nature of asset performance in ESG portfolios. This study extends the growing body of research on ESG adoption in ASEAN + 6 and its implications for investors and policymakers. South Korea and Indonesia highlight the challenges of balancing growth and sustainability at different stages of market development, providing valuable insights into how the different assets analysed can serve as hedging and diversification tools for SRI investment during periods of global financial disruption.

The paper is set out as follows. Section two reviews the literature and section three presents the empirical methodology. Section four describes the data, while section five provides the empirical findings, which focus on conditional correlations, portfolio management and the effectiveness of the given hedge ratios and diversification. Section six contains the discussion of our findings and section seven highlights the main conclusions and what these imply for investors and financial policymakers.

Literature review

Previous literature has explored the importance of a robust governance framework, technological innovation, and organizational flexibility in the adoption of sustainability goals (Shabbir et al., 2020, Liu et al., 2021, Li et al., 2022) and their relationship with a firm's financial performance (Shabbir and Wisdom, 2020).

In this regard, extensive literature has explored the performance of SRI indices globally as ESG principles have been increasingly incorporated by investors into their decision-making

processes. The performance of SRI indices varies depending on whether they correspond to developed or emerging markets. Bodhanwala and Bodhanwala (2020) showed that SRI portfolios in developed economies like the United States and Australia perform similarly to conventional indices. However, in emerging markets, SRI portfolios often underperform due to smaller sample sizes and limited diversification. Shabbir and Muhammad (2019) emphasized the importance of foreign portfolio investments in improving market liquidity and reducing volatility, which is crucial for markets like Indonesia.

Another branch of the literature focuses on diversification opportunities and hedging strategies by analyzing historical correlations between equity investments and commodities or fixed-income products. The increasingly interconnected and unpredictable nature of financial markets, influenced by global events, requires accurate modelling and forecasting of volatility. These tools are critical for pricing derivatives, optimizing portfolios, managing risk and implementing effective hedging strategies. The urgency to incorporate additional criteria, particularly SRI, has made this area of study increasingly relevant in today's financial landscape.

Despite growing interest, the literature combining SRI indices with commodities to build effective portfolios and manage risks remains limited (Dutta et al. 2020; Cupriak et al. 2020; Shahid et al. 2023). Most existing studies have primarily focused on developed markets (Roca et al. 2010; Tularam et al. 2010; Sadorsky 2014; Giannarakis et al. 2016; Miralles-Quirós, Miralles-Quirós (2017); Cupriak et al. 2020; and Shahid et al. 2023), leaving a gap in understanding the dynamics of SRI portfolios in emerging economies, such as Indonesia, which face unique regulatory and economic challenges. South Korea, a developed market, offers a valuable comparison, demonstrating advanced ESG integration and resilience during financial crises.

The combination of SRI indices with commodities, green bonds, and traditional financial instruments presents significant diversification opportunities. De Oliveira et al. (2017) examined the impact of oil prices on the Brazilian Corporate Sustainability Index (ISE), finding that crude oil spot prices, especially WTI, significantly influence the behavior of socially responsible companies. This aligns with Iglesias-Casal et al. (2020), who studied diversification potential using multivariate GARCH models and found that gold and the Oil Volatility Index (OVX) offer the greatest benefits for reducing portfolio risk when combined with SRI indices. These findings emphasize the value of commodities in SRI portfolio construction, particularly in emerging markets.

Gold has consistently been highlighted as an effective tool for diversification. Mensi et al. (2017) provided evidence of time-variant risk spillovers between crude oil, gold, Islamic stock indices, and sustainability indices, showing that gold is more advantageous in terms of portfolio diversification and downside risk reduction. Green bonds have also emerged as critical assets for diversification. Nguyen et al. (2021) and Mensi et al. (2022) found that green bonds exhibit low or negative correlations with stocks and commodities, making them effective tools for diversification during crises such as the COVID-19 pandemic. Reboredo (2018) and Liu et al. (2021) further emphasized the asymmetric spillover effects of green bonds, particularly during extreme market conditions, highlighting their role in multi-asset portfolios.

Specifically, studies on the diversification potential of SRI investments in Indonesia and South Korea remain scarce. Robiyanto et al. (2021) using a DCC-GARCH model with monthly data from 2013 to 2019 for the SRI-KEHATI Index and Jakarta Islamic Index, confirmed that combining gold with SRI reduces risk compared to individual assets. However, the risk-adjusted return of such a portfolio does not exceed that of SRI.

Robiyanto et al. (2023a) expanded the analysis of the SRI-KEHATI index by adding gold, cryptocurrencies and corporate bonds using daily data from 2019 to 2021 and DCC-GARCH analysis. Their findings revealed that adding Ripple or gold to SRI portfolios significantly reduces investment risk. Similarly, Asih et al. (2024) applied a DCC-GARCH methodology with daily return from 2018 to 2023, showing that ESG-based equities in Indonesia, represented by ETF SRI-KEHATI, act as effective diversification tools for other ETFs, particularly during periods of market turbulence.

Regarding South Korea, Ang (2015), using daily data from 2006 to 2014 with a univariate GARCH, concluded that the DJSKI provides stable returns and low sensitivity to market downturns, making it an attractive option for international diversification. Ang and Weber (2017) further examine the hedge and safe haven properties of gold relative to socially responsible and conventional investment in Korea using an ARDL model. Their findings suggest that gold's role as a strong diversifier for socially responsible and conventional investment.

Despite these insights, no studies to date have applied the Asymmetric Dynamic Conditional Correlation (A-DCC) model with GJR-GARCH in conditional variance and out-of-sample analysis to explore dynamic conditional correlations, optimal weights, and hedge ratios for SRI portfolios in South Korea. The unique economic and regulatory contexts of South Korea remain underexplored, particularly their implications for portfolio optimization and risk management.

In the case of Indonesia, our contribution relative to existing studies lies in constructing hedging portfolios for socially responsible investment and comparing them with optimally weighted portfolios using the hedge effectiveness (HE) ratio.

Another novel aspect in this context is the calculation of optimal weights and hedge ratios through an out-of-sample analysis based on an A-DCC model estimated with rolling windows. Furthermore, our study integrates assets not previously used in similar research, including a commodity futures index, U.S. Treasury bond futures, and a green bond index. The analysis also covers two major global crises: the COVID-19 pandemic and the geopolitical turmoil resulting from the Russia-Ukraine war.

With this in mind, the following section presents the paper's methodology.

Methodology

For portfolio diversification strategies, analyzing the performance and risk of various assets is essential. To examine the dynamic evolution of an asset's risk, univariate GARCH models can be employed, where current volatility depends on past shocks and volatilities. When working with daily data, these models effectively capture day-to-day fluctuations in an asset's volatility.

Equally important is the estimation of the dynamic correlation structure between asset returns. A wide range of multivariate GARCH models exists for this purpose, each with distinct characteristics. Among these models, the Dynamic Conditional Correlation (DCC) model and its asymmetric extension, the A-DCC model, are widely used due to their flexibility. This study employs the A-DCC model, which, to the best of our knowledge, has not been previously applied in research on Indonesia and South Korea. These models are highly flexible, enabling the direct estimation of dynamic conditional correlations with a reduced number of parameters compared to other models like BEKK or asymmetric BEKK (A-BEKK). Moreover, these parameters have direct interpretability. Combining conditional variances and correlations allows for the calculation of the weight's assets will receive in a portfolio, thereby providing measures of performance and associated risk reduction for the constructed portfolio.

Section “A-DCC model” describes the A-DCC model used to obtain dynamic conditional correlations and the GJR-GARCH model used to estimate the conditional variances of the different assets. Based on these results, the hedge ratios described in Section “Dynamic hedge ratios” and the optimal portfolio weights (Section “Dynamic portfolio weights”) are calculated. Finally, Section “Hedging effectiveness” provides a measure of portfolio risk reduction or hedge effectiveness.

A-DCC model. Multivariate GARCH models describe the dynamic co-movements that arise among different market assets. To this end, our paper uses the A-DCC MGARCH model proposed by Cappiello et al. (2006), which is widely used in financial literature.

The equations of this model are defined as follows:

$$Y_t = \mu_t + \varepsilon_t \quad \varepsilon_t = H_t^{\frac{1}{2}} z_t \quad (1)$$

$$H_t = D_t R_t D_t \quad (2)$$

$$\varepsilon_t / \Omega_{t-1} \rightarrow \text{Student} - t(0, H_t, \nu)$$

where Y_t is a 4×1 returns vector, μ_t is a 4×1 vector measuring the conditional mean of Y_t and which may contain constants or a stochastic process ARMA(p, q); H_t is the conditional covariance matrix of the residuals ε_t ; Ω_{t-1} is an information set available in time $t-1$; z_t a vector of the error terms which are identically and independently distributed (i.i.d.) as a t-Student of ν degrees of freedom ($2 < \nu < \infty$), an average of zero $E[z_t] = 0$ and $E[z_t z_t'] = I$; and D_t the diagonal matrix of the conditional standard deviations.

The equations for the conditional variance of each element of Y_t are given by the following GJR-GARCH(1,1) process proposed by Glosten et al., (1993):

$$h_{i,t} = \omega_i + \alpha_i \varepsilon_{i,t-1}^2 + \lambda_i \varepsilon_{i,t-1}^2 I(\varepsilon_{i,t-1}) + \beta_i h_{i,t-1} \quad (3)$$

where the function indicator $I(\varepsilon_{i,t-1})$ is equal to 1 if $\varepsilon_{i,t-1} < 0$ and 0 otherwise. This allows for the possibility that bad news may increase volatility to a greater extent than good news.

Finally, the matrix of conditional correlations R_t is defined as follows:

$$R_t = (\text{diag}(Q_t))^{-\frac{1}{2}} Q_t (\text{diag}(Q_t))^{-\frac{1}{2}} \quad (4)$$

where:

$$Q_t = ((1 - \theta_1 - \theta_2)P - \theta_3 N) + \theta_1 z_{t-1} z_{t-1}' + \theta_3 \eta_{t-1} \eta_{t-1}' + \theta_2 Q_{t-1} \quad (5)$$

in which θ_1 , θ_2 and θ_3 are parameters, P is the matrix of the unconditional covariances of the standardized residuals and $N = E(\eta_{t-1} \eta_{t-1}')$ being $\eta_{t-1} = I(z_{t-1} < 0) \varepsilon_{t-1}$, where $I(\cdot)$ is an indicator function and the operator $\circ$ denotes the Hadamard product.

The estimation of the parameters θ_1 , θ_2 and θ_3 for quasi-maximum likelihood is obtained maximizing the following log-likelihood function:

$$L(\theta) = \sum_{t=1}^T \left(\ln \Gamma\left(\frac{\nu+m}{2}\right) - \ln \Gamma\left(\frac{\nu}{2}\right) - \frac{m}{2} \ln((\nu-2)\pi) - \frac{1}{2} \ln |R_t| - \ln |D_t|^{1/2} \right) - \frac{\nu+m}{2} \sum_{t=1}^T \ln \left(1 + \frac{\varepsilon_t' R_t^{-1} \varepsilon_t}{\nu-2} \right) \quad (6)$$

where m is the number of elements of the vector Y_t , i.e. 4.

This methodology describes the dynamic behaviour of volatilities and correlations, and the role of the negative shocks during crisis periods compared to calm periods.

Dynamic hedge ratios. One hedge strategy for efficiently managing portfolios is by calculating the optimal hedge ratio. This ratio allows an investment of one dollar in a particular market to be hedged against tail risks by selling a specific amount in another futures equity or commodity market, aiming to minimize risk.

The optimal hedge ratio, $\beta_{ij,t}$, introduced by Kroner and Sultan (1993), is defined for a two-asset portfolio as follows.

$$\beta_{ij,t} = \frac{h_{ij,t}}{h_{j,t}} \quad (7)$$

Where $h_{ij,t}$ is the conditional covariance of asset i and j , and $h_{j,t}$ is the conditional variance of asset j . This definition indicates that a long position of one dollar in asset i can be hedged by taking a short position of $\beta_{ij,t}$ in asset j . The portfolio based on this approach is especially useful when hedging a spot position using futures.

Dynamic portfolio weights. The optimal holding weight of asset j in a portfolio of one dollar containing two assets, i and j for the period t ($w_{j,t}$), is described by the following expressions (Kroner and Ng (1998)):

$$w_{j,t} = \frac{h_{i,t} - h_{ij,t}}{h_{j,t} - 2h_{ij,t} + h_{i,t}} \quad (8)$$

and,

$$w_{j,t} = \begin{cases} 0, & \text{if } w_{j,t} < 0 \\ w_{j,t} & \text{if } 0 \leq w_{j,t} \leq 1 \\ 1, & \text{if } w_{j,t} > 1 \end{cases}$$

where $h_{i,t}$ and $h_{j,t}$ are the conditional variance of asset i and j respectively, $h_{ij,t}$ is the conditional covariance between the two assets in the period t . Therefore, the optimum weight of the i asset is given by $1 - w_{j,t}$.

In the following, we discuss two strategies used in our analysis in search of the most efficient portfolio management as well as how to calculate the effectiveness of the strategies used.

Hedging effectiveness. In this section, we calculate the effectiveness of portfolio strategies. For this purpose, the portfolio containing only asset i is used as a benchmark to compare two portfolios: a hedging portfolio and a portfolio of optimal weights. The returns of these portfolios are given, respectively, by the following expressions.

$$r_{\beta,t} = y_{i,t} - \beta_{ij,t} y_{j,t} \quad (9)$$

$$r_{w,t} = (1 - w_{j,t}) y_{i,t} + w_{j,t} y_{j,t} \quad (10)$$

The portfolio's effectiveness in reducing risk is assessed by comparing the benchmark portfolio with those described above. The effectiveness of the hedge is measured by the hedge effectiveness index (HE), which may be obtained, following Ku et al. (2007) and Chang et al. (2011), as:

$$HE = \left(\frac{\text{Var}_{unhedged} - \text{Var}_{hedged}}{\text{Var}_{unhedged}} \right) \quad (11)$$

Where Var_{hedged} is the variance obtained either from the hedging portfolio or the optimal weights portfolio, while $\text{Var}_{unhedged}$ is the variance of individual asset return i . A high HE value indicates a greater reduction in portfolio variance and consequently entails enhanced risk mitigation. The comparison between the two portfolios provides valuable information for investors, offering them a practical decision-making tool.

Table 1 Descriptive statistics of daily returns.

	RDJSKI	RJSRI	RGI	RGOLD	RTBs	RGBs	RKS200	RJKSE
Mean	0.0030	0.0092	0.0031	0.0237	−0.0057	−0.0032	0.0078	0.0053
Median	0.0000	0.0000	0.0238	0.0000	0.0000	0.0002	0.0000	0.0000
Maximum	12.3406	15.7982	7.5959	5.7775	1.9727	2.6301	11.8911	10.7737
Minimum	−10.2924	−11.5675	−12.7625	−5.1069	−1.4592	−3.0305	−8.0894	−9.8438
Std. Dev.	1.3503	1.3898	1.4500	0.9067	0.3376	0.4372	1.2899	1.1261
Skewness	0.0331	−0.05888	−0.7467	−0.0254	0.2173	−0.1103	0.0226	−0.4970
Kurtosis	10.3575	17.4676	10.9111	7.1611	5.6767	7.2508	8.7870	14.7893
Jarque-Bera	5567.14***	21525.7***	6665.21***	1780.83***	756.244***	1863.13***	3443.99***	14394.27***
ADF test	−31.46***	−45.98***	−48.70***	−50.89***	−51.12***	−44.88***	−31.92***	−46.09***
Q10(r_t)	41.270***	20.850**	10.626	14.604	14.929	33.314***	33.539***	26.139***
Q10(r_t^2)	2521.3***	1459.4***	594.30***	241.63***	605.35***	715.14***	1687.9***	1988.7***
Corr. RDJSKI			0.239***	0.120***	−0.045***	0.227***	0.966***	
Corr. RJSRI			0.150***	0.020	−0.080***	0.108***		0.955***
Observations	2468	2468	2468	2468	2468		2468	2468

Source: Own elaboration, Datastream, Bloomberg and S&P Global.

Jarque Bera denotes the Jarque-Bera normality test; ADF is the Augmented Dickey-Fuller unit root test; Q10(r_t^2) is the Ljung-Box Q-statistic of order 10 for the squares of the returns; symbols **, *** indicate the rejection of null hypothesis at the 5% and 1% levels, respectively.

Data

The empirical investigation is based on daily observations of the JKSRI (Indonesia SE Sri-Kehati Index), the DJSKI (Dow Jones Sustainability Korea Index), the Jakarta Composite Index (JKSE) and the Korea Stock Exchange KOSPI200 (KS200). The data cover the period from 14th October 2014 to 29st March 2024 (2,565 observations). The selection of this sample period is aligned with the availability of data for one of the indices used.

In addition to the equity indices, we also consider gold, using the price series of the COMEX Gold Composite Commodity Future (GOLD), and a basket of commodities represented by the Goldman Sachs CI Electronic Commodity Index Future (GI) on the Chicago Mercantile Exchange. The GI is a globally recognized commodity index that tracks 24 exchange-traded futures contracts tied to physical commodities across five sectors: energy, industrial metals, precious metals, agriculture and livestock. In 2021, energy accounted for the largest share at 53.9%, while precious metals comprised the smallest at 6.9% of the total index composition. The Barclays MSCI Green Bonds Index (GBs) was selected as a measure of the global GB market. This index, launched in the fourth quarter of 2014, includes investment grade-rated green bonds with fixed-rate coupons issued in multiple currencies by corporations, government entities and securitized bonds. These bonds are categorized as green by MSCI ESG Research, covering sectors such alternative energies, energy efficiency, pollution prevention, sustainable water, green building and climate adaption, using criteria aligned with the Green Bond Principles (GBP). The index is rebalanced monthly, and bonds remain included until they reach maturity. CBoT 10-Year US Treasury Bond Futures (TBs) were also included in the analysis.

The DJSKI (Dow Jones Sustainability Korea Index) data are sourced from S&P Global, while the Barclays MSCI Green Bonds Index data come from Bloomberg. All other data series are taken from Datastream. All series are expressed in US dollars. The daily returns series $y_{i,t}$ are calculated as $100 \cdot \ln(P_{i,t}/P_{i,t-1})$, where $P_{i,t}$ is the daily closing price. Descriptive statistics and stochastic properties for the return data for DJSKI, JKSRI, GI, Gold, TBs, GBs, JKSE and KS200 are reported in Table 1, while the evolution of the price data is presented in Fig. 1.

As evidenced in Table 1, all return series are stationary (ADF test), as the null hypothesis of a unit root is rejected, and they are characterized by the absence of normality as indicated by the Jarque-Bera test. During the analyzed period, each of the sustainable and market indices shows a positive mean return. In the

case of Indonesia, unlike in South Korea, the mean return of the sustainable index exceeds that of the market index. The rest of the assets considered also exhibit positive mean returns, except for treasury bonds and GBs. Gold has the highest mean return among the assets. The standard deviations range from a minimum of 0.36 (TBs) to a maximum of 1.45 (GI) and the DJSKI return has a lower standard deviation than the JKSRI. The high kurtosis values and negative skewness values indicate a higher probability of extreme events and losses, respectively. The Ljung-Box statistic reveals that all returns exhibit autocorrelation except for GI, Gold, and TBs. The presence of autocorrelation in the squared returns of all assets indicates the existence of conditional heteroscedasticity.

The two sustainable indices are highly correlated with the respective market index but exhibit relatively low, albeit significant, correlations with the other assets used to construct the portfolios. They show positive correlations with GI, gold and GBs, and negative correlations with treasury bonds. The two sustainable indices have the highest correlation with GI. The weakest correlation for the Indonesian sustainable index is with gold, whereas treasury bonds have the lowest correlation with the Korean sustainable index.

Figure 1 shows that the sustainable, market and commodity indices experienced a significant fall in March 2020 at the beginning of the COVID-19 pandemic, followed by a subsequent recovery, which was more pronounced for the DJSKI and KS200. In fact, at the beginning of 2021, the DJSKI reached an all-time peak in quoted values. Conversely, the GI saw a notable increase during the 2020–2022 period, coinciding with the rising oil prices. However, TBs and GBs showed significant declines in 2021 and specially in 2022 due to consecutive interest rate hikes in the US. Unlike the other indices, gold, in its role as a safe haven, did not experience any decrease in the prices during the analyzed period.

Empirical findings

This section examines the results of portfolios constructed as an alternative to individual investment in each sustainable index. First, we present the results of the estimated A-DCC models with five variables for each country. These are divided into three parts: equations of the mean, equations of the conditional variance and an equation of the estimated conditional correlations, which considers a multivariate t distribution. Second, we discuss the results regarding conditional correlations, optimal hedging

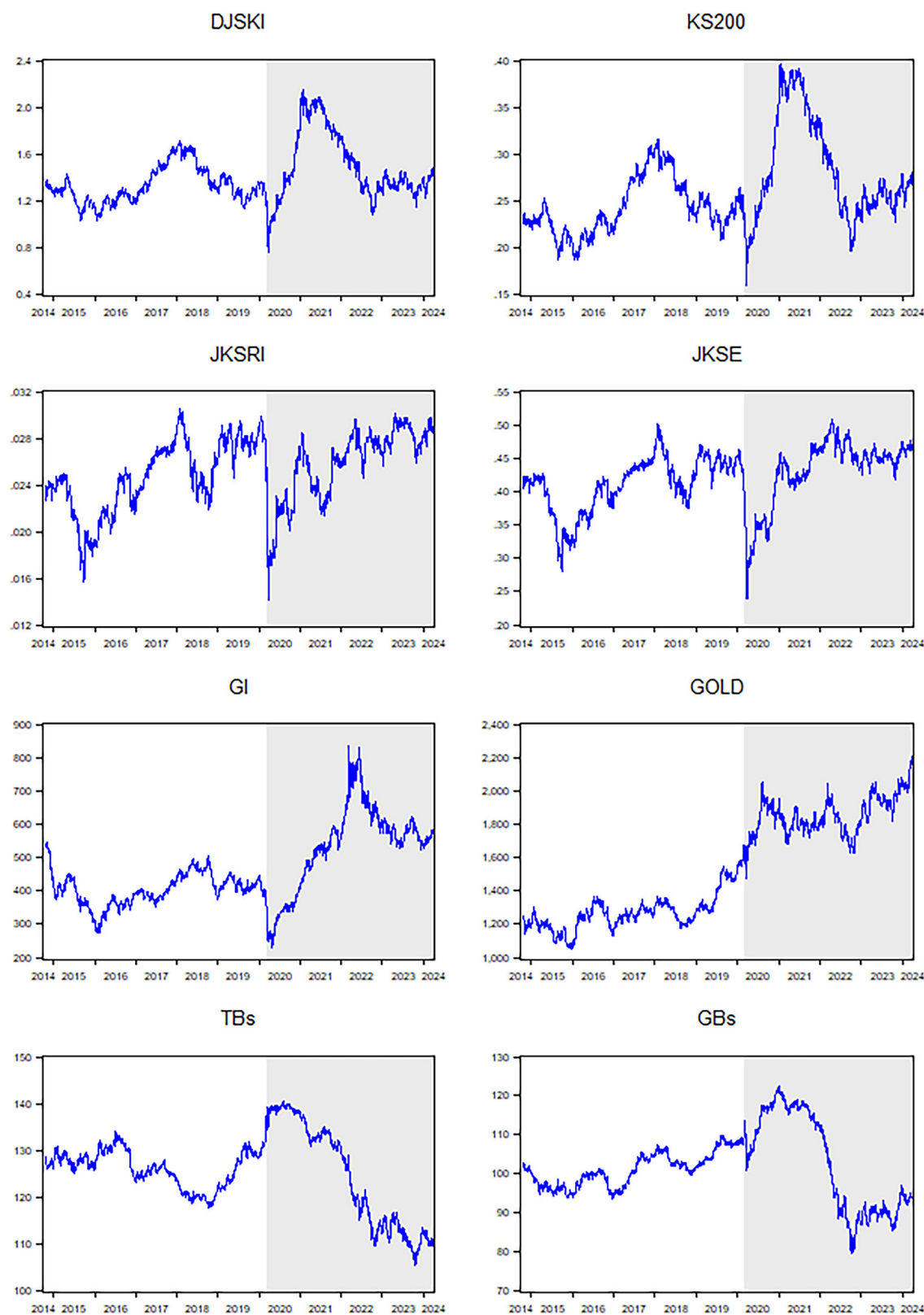


Fig. 1 Daily stocks prices. Source: Datastream, Bloomberg and S&P Global. The shaded area corresponds to the period of the COVID-19 pandemic and war period.

ratios, optimal portfolio weights and the effectiveness of these new portfolios in mitigating risk compared to individual investment.

The A-DCC model estimates for Indonesia are provided in Table 2 below. In the mean equations, an AR(1) structure was

included. Notably, the coefficient at lag 1 (φ_1) is significant in all equations except in the GI equation.

In the variance equation, which includes a GJR-GARCH(1,1) model, it is observed that long-term persistence β surpasses short-term persistence α , which is in line with expectations. The

Table 2 A-DCC parameter estimates.

	JKSRI	GI	GOLD	TBs	GBs
Mean equation					
μ	0.0131	0.0575**	0.0292**	−0.0039	0.0045
φ_1	−0.0012*	0.0267	−0.0363**	−0.0450**	0.0473**
Variance equation					
ω	0.0214*	0.0386***	0.0091***	0.0008*	0.0012***
α	0.0162	0.0319**	0.0404***	0.0599***	0.0395***
β	0.9352***	0.9167***	0.9636***	0.9678***	0.9505***
λ	0.0739***	0.0605***	−0.0228*	−0.0066	0.0072
ν	4.8204***	5.0796***	3.7170***	7.2700***	6.9612***
Correlation equation					
θ_1	0.0157***				
θ_2	0.9462***				
θ_3	0.0000				
$m\nu$	6.3961***				
Ln L	−11515				
AIC	9.3714				
SBC	9.4868				
Obs.	2468				

Indonesia SRI.

Symbols ***, **, * indicate significance at the 1%, 5%, and 10% levels, respectively. Ln L, AIC and SBC are the log-likelihood, the Akaike information criterion and the Schwarz Bayesian information criterion values respectively. The data used for this model estimation corresponds to the period between October 15, 2014, and March 29, 2024.

Table 3 A-DCC parameter estimates.

	DJSKI	GI	GOLD	TBs	GBs
Mean equation					
μ	0.0067	0.0575**	0.0292**	−0.0039	0.0045
φ_1	0.0729***	0.0267	−0.0363**	−0.0449**	0.04728**
Variance equation					
ω	0.0535	0.0386***	0.0091***	0.0008*	0.0011***
α	0.0319*	0.0319**	0.0404***	0.0599***	0.039***
β	0.9053***	0.9167***	0.9636***	0.9377***	0.9504**
λ	0.0575**	0.0604***	−0.0228*	−0.0066	0.0071
ν	7.6286***	5.0795***	3.7169***	7.2700***	6.9611***
Correlation equation					
θ_1	0.0167***				
θ_2	0.9536***				
θ_3	0.0000				
$m\nu$	6.8129***				
Ln L	−11565				
AIC	9.4117				
SBC	9.5271				
Obs.	2468				

Korea SRI.

Symbols ***, **, * indicate significance at the 1%, 5%, and 10% levels, respectively. Ln L, AIC and SBC are the log-likelihood, the Akaike information criterion and the Schwarz Bayesian information criterion values respectively. The data used for this model estimation corresponds to the period between October 15, 2014, and March 29, 2024.

statistical significance of both terms, except α in the JKSRI equation, confirms the existence of volatility clustering. Notably, the estimated asymmetric coefficient (λ) is positive and statistically significant for JKSRI and GI, suggesting that adverse events tend to increase volatility more than positive shocks of the same magnitude. Conversely, these coefficients are negative and statistically significant for gold, indicating that negative shocks tend to reduce volatility for these returns. The stability condition is fulfilled for the variance equations.

For the conditional correlation equation, the estimated coefficients θ_1 and θ_2 are positive and statistically significant and the dynamic conditional correlation is mean reverting. The degrees of freedom for the univariate and multivariate Student's t-distribution (ν and $m\nu$) are significant, suggesting that the distribution of returns exhibits fat tails.

In the results of the A-DCC model estimation for Korea, shown in Table 3, similar behaviour is observed in the equations of the means and dynamic conditional correlations as in the Indonesian model. The main difference lies in the variance equations, where the estimated α coefficients are significant.

The aforementioned estimates, where the sustainable indices are substituted for Indonesian and Korean capital market indices (JKSE and KS200, respectively), are presented in Tables A.1 and A.2 in the appendix. There is very little divergence in the performance of both types of indices².

In the next phase of the study, we use an out-of-sample analysis to obtain dynamic conditional variances and correlations, hedging coefficients and optimal portfolio weights. Specifically, a rolling estimation of the A-DCC models with a fixed window size is used and refitted every 20 observations. The first estimation is

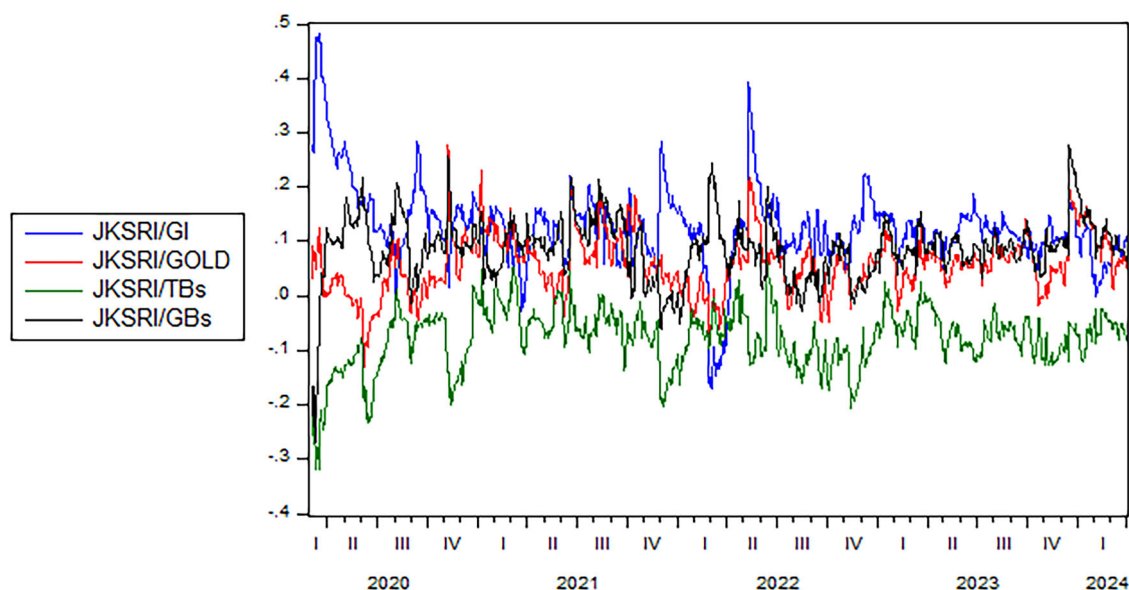


Fig. 2 Rolling one-step-ahead dynamic conditional correlations. Indonesia.

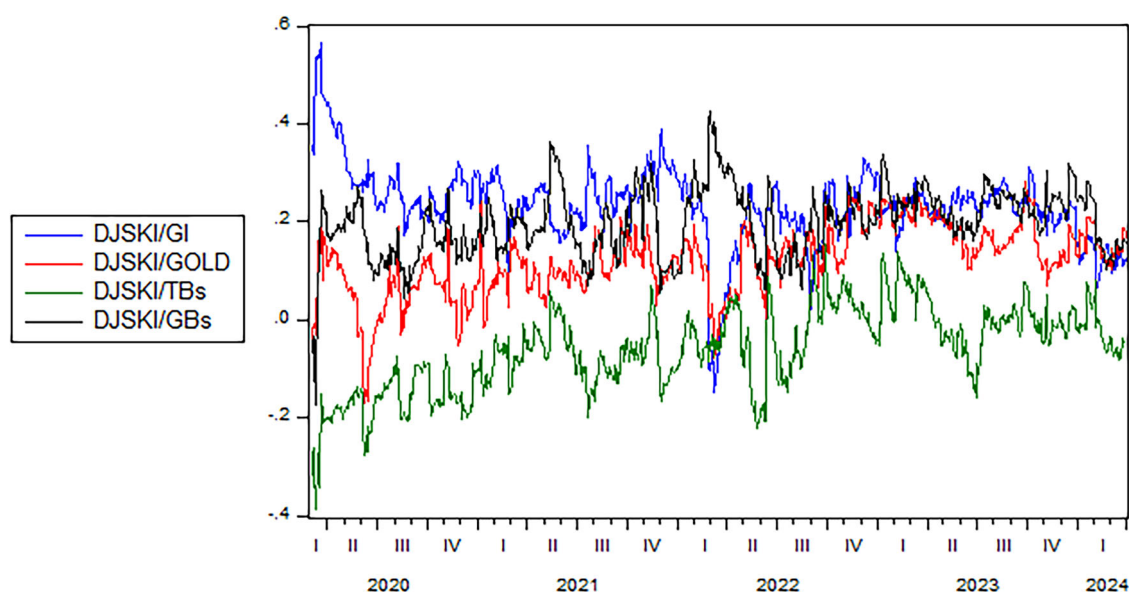


Fig. 3 Rolling one-step-ahead dynamic conditional correlations. Korea.

carried out using data from the start of our analysis until March 2nd, 2020, providing fixed windows of 1,404 observations. With rolling estimates of this size, 1,064 one-step-ahead forecasts are made for the period from 3-3-2020 to 3-29-2024. Given the impact of the Ukraine War on the global economy, the analysis is divided into two distinct periods, February 24th, 2022 making the start of the second period.

Figures 2 and 3 show the one-step-ahead pairwise dynamic conditional correlations for Indonesia and Korea. In the case of Indonesia, the correlation of the sustainable index with the GI is positive throughout most of the period and, on average, is the highest among the four assets considered, followed by GBs. A similar pattern is observed with gold, although the negative correlations persist over a longer period. However, the average correlation with bonds is negative. For Korea, the correlation of the sustainable index with the other assets mirrors the signs observed for Indonesia but with higher values for the positive correlations.

The hedge ratios are computed in the same way as the dynamic conditional correlations, i.e. 1,064 out-of-sample forecasts are made available from the rolling window analysis. Figure 4 illustrates the chronological evolution of the hedge ratios for Indonesia, while Table 4 (Panel A) provides descriptive statistics and the hedging effectiveness. Hedge ratios vary across the sample, highlighting the need for frequent updates. The hedge ratios calculated for the TBs exhibit the highest levels of variability.

In Indonesia, the average hedge value for GBs (Table 4 Panel A) stood at 0.26 during the full rolling sample, which is considerably higher than that for GI (0.12) and gold (0.07). This suggests that the acquisition of one dollar of JKSRI may be hedged by a short position of 0.26 dollars in the GBs index. Conversely, the hedge ratio of the treasury bonds is negative (-0.33), which is a consequence of negative conditional correlations and indicates that long positions in JKSRI and bonds should be taken. When the Indonesian market index is pairwise

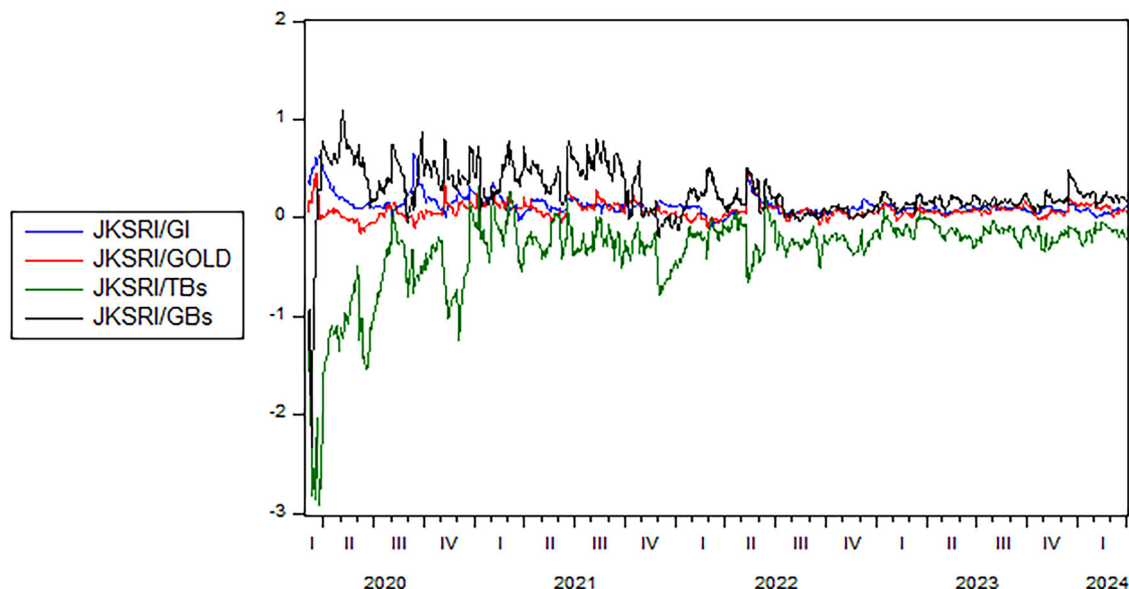


Fig. 4 Rolling one-step-ahead hedge ratios. Indonesia. The hedge ratios refer to GI, Gold, TBs and GBs in the portfolios considered.

Table 4 Hedge ratios and portfolio weights: summary statistics and hedging effectiveness.

	Full rolling sample ($n = 1064$)				Pre-WAR ($n = 517$)				WAR ($n = 547$)			
	(3/03/2020-3/29/2024)				(3/03/2020-2/23/2022)				(2/24/2022-3/29/2024)			
Panel A: Hedge ratios												
	Mean	Min	Max	HE	Mean	Min	Max	HE	Mean	Min	Max	HE
JKSRI/GI	0.1223	−0.1122	0.6474	0.0333	0.1651	−0.0236	0.6474	0.0414	0.0818	−0.1122	0.4185	−0.0030
JKSRI/GOLD	0.0739	−0.1538	0.5024	−0.0257	0.0775	−0.1538	0.4528	−0.0322	0.0706	−0.0976	0.5024	0.0036
JKSRI/TBs	−0.3351	−2.9262	0.3159	−0.0042	−0.5030	−2.9262	0.3159	−0.0038	−0.1764	−0.6474	0.1840	−0.0061
JKSRI/GBs	0.2561	−2.3354	1.0939	−0.0256	0.3564	−2.3354	1.0939	−0.0256	0.1613	−0.0350	0.5088	0.0058
Panel B: Portfolio weights												
	Mean	Min	Max	HE	Mean	Min	Max	HE	Mean	Min	Max	HE
JKSRI/GI	0.5736	0.0714	0.9155	0.1635	0.4706	0.0714	0.8393	0.3360	0.6711	0.3514	0.9155	−0.6031
JKSRI/GOLD	0.4002	0.0234	0.6589	0.3357	0.3390	0.0234	0.6042	0.3124	0.4582	0.0919	0.6589	0.4398
JKSRI/tBs	0.1209	0.0094	0.3989	0.1262	0.0483	0.0094	0.1380	0.0789	0.1896	0.0480	0.3988	0.3376
JKSRI/GBs	0.1421	0.0000	0.5229	0.1081	0.0435	0.0000	0.1609	0.0562	0.2353	0.0205	0.5229	0.3398

Indonesia Sustainable Index.
Both the optimal weightings and the optimal hedge ratios refer to the second asset in the portfolios considered.

combined with the four assets, the results are similar (Table A3 Panel A).

For Korea, the data shown in Fig. 5 and Table 5 Panel A also reveal significant variation over the analyzed period. Furthermore, the average hedge ratio for GBs was 0.60 during the full rolling sample, for GI 0.23, gold 0.17 and treasury bonds -0.32. While the hedge ratios differ from those in Indonesia, the same general behaviour is observed. The KS200 market index analysis provides hedge ratios similar to those of the DJSKI sustainable index (Table A4 Panel A).

It is also worth noting that the cost of hedging in both countries varies significantly when comparing the war period and the one that precedes it.

An alternative diversification strategy involves the optimal weights portfolio. Figures 6 and 7, for Indonesia and Korea respectively, show the evolution of optimal weights assigned to the asset combined with the sustainable index in a two-asset portfolio. As with hedge ratios, optimal weights fluctuate significantly over time, except for TBs and GBs, whose weighting does not exhibit excessive variability. This underscores the need for an active strategy in allocating portfolio weights.

The average weights, as presented in Table 4 Panel B for Indonesia and Table 5 Panel B for Korea, reveal that commodities receive the highest portfolio weightings for the two periods analysed. In Indonesia, 57.3 cents should be invested in GI and 42.7 cents in JKSRI in a one-dollar JKSRI/GI portfolio. In Korea, 52.7 cents should be invested in GI and 47.3 cents in DJSKI in a one-dollar DJSKI/GI portfolio. During the war period, all assets received higher weights when combined with sustainable indices in Indonesia. However, in Korea the weight of gold decreased during the war period.

When the sustainable index is replaced by the market index in Indonesia, the optimal weights for the four assets increase (Table A3 Panel B). For example, in the JKSE/GI pair, 66.2 cents should be invested in GI compared to 33.8 cents in the JKSRI/GI portfolio. The same pattern is observed in the case of Korea, (Table A4 Panel B).

The baseline for determining the optimal investment strategy is established through an assessment of effective risk reduction within the portfolio. To assess hedging effectiveness (HE), a thorough analysis is conducted, with results presented in Tables 4 and 5 for portfolios comprising sustainable indices, and in Tables A3 and A4 for portfolios comprising market indices.

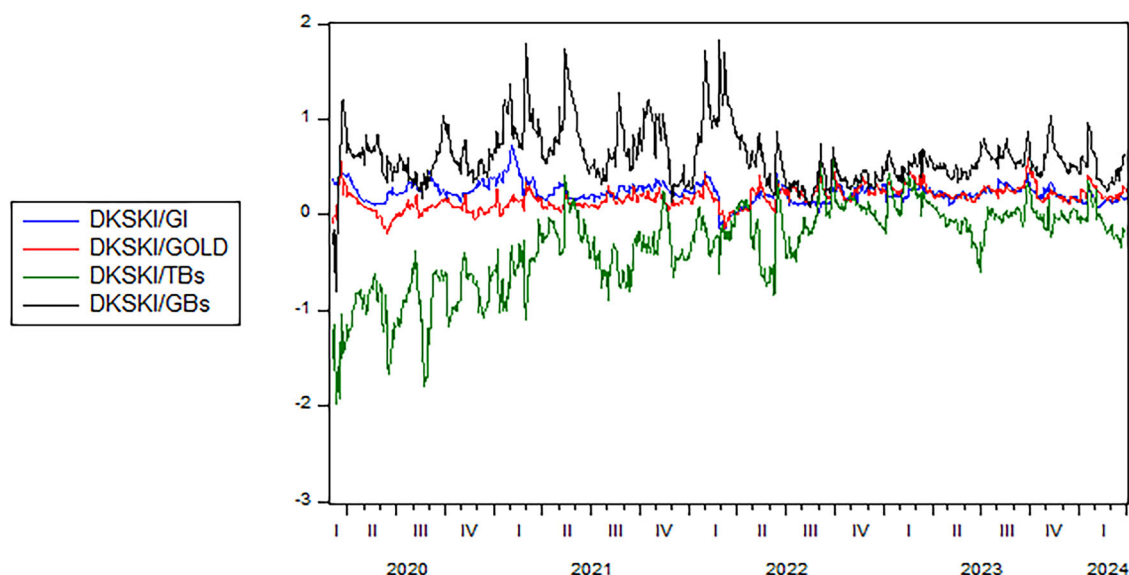


Fig. 5 Rolling one-step-ahead hedge ratios. Korea. The hedge ratios refer to GI, Gold, TBs and GBs in the portfolios considered.

Table 5 Hedge ratios and portfolio weights: summary statistics and hedging effectiveness.

	Full rolling sample (<i>n</i> = 1064)				Pre-WAR (<i>n</i> = 517)				WAR (<i>n</i> = 547)			
	(3/03/2020-3/29/2024)				(3/03/2020-2/23/2022)				(2/24/2022-3/29/2024)			
Panel A: Hedge ratios												
	Mean	Min	Max	HE	Mean	Min	Max	HE	Mean	Min	Max	HE
DJSKI/GI	0.2309	−0.2105	0.7225	0.0733	0.2724	0.1058	0.7225	0.1032	0.1916	−0.2105	0.4356	0.0215
DJSKI/GOLD	0.1729	−0.1912	0.6091	0.0049	0.1187	−0.1912	0.5657	−0.0116	0.2241	−0.1476	0.6091	0.0331
DJSKI/TBs	−0.3188	−1.9780	0.6774	−0.0050	−0.6115	−1.9780	0.4054	0.0008	−0.0421	−0.8457	0.6774	−0.0152
DJSKI/GBs	0.6033	−0.8097	1.8190	0.0614	0.6870	−0.8097	1.7817	0.0568	0.5242	0.0812	1.8190	0.0689
Panel B: Portfolio weights												
	Mean	Min	Max	HE	Mean	Min	Max	HE	Mean	Min	Max	HE
DJSKI/GI	0.5274	0.0550	0.9877	0.1545	0.5131	0.0550	0.9877	0.2233	0.5409	0.1486	0.8514	0.0218
DJSKI/GOLD	0.3350	0.0496	0.5944	0.5057	0.3603	0.0496	0.5944	0.4956	0.3112	0.0576	0.5921	0.5252
DJSKI/TBs	0.0853	0.0113	0.2968	0.4061	0.0559	0.0113	0.1258	0.3846	0.1131	0.0138	0.2968	0.4487
DJSKI/GBs	0.0687	0.0000	0.4237	0.3119	0.02830	0.0000	0.1089	0.2812	0.1070	0.0000	0.4238	0.3727

Korea Sustainable Index.
Both the optimal weightings and the optimal hedge ratios refer to the second asset in the portfolios considered.

In Indonesia, hedging provides a small reduction in risk only for commodities over the full rolling sample and pre-war period, with hedge effectiveness values of 0.033 and 0.041, respectively. During the war period, portfolios containing GBs or gold showed small reductions in risk. In Korea, except for TBs during the full rolling period and the war period, most assets reduced risk to some extent. However, in the pre-war period, gold was the only asset that did not reduce portfolio risk. Generally, GI and GBs were the most effective hedging instruments.

In short, in Indonesia, when using the hedging ratio strategy, GI was effective in reducing the level of risk for socially responsible investment over the full rolling period. In Korea, the largest reduction was observed for GI and GBs. In both countries, GI provided the greatest risk reduction over the full period, while GBs were more effective in the last sub-period. A similar pattern was observed for the market indices.

In optimally weighted portfolios, the variance reduction consistently surpasses that of the hedged portfolios. The only exception was the portfolio containing commodities in Indonesia during the war period, which showed a negative HE.

Overall, in optimal weighting portfolios with sustainable indices, a risk reduction is achieved for all four assets considered,

implying diversification opportunities. The greatest risk reduction is obtained in portfolios that include gold, except for Indonesia in the pre-war period. Our conclusions are similar to the findings of Mensi et al. (2017) and Iglesias-Casal et al. (2020).

In Indonesia (as indicated in Table 4 Panel B), the greatest risk reduction was observed in the portfolio containing gold (0.34 for the full rolling sample and 0.44 for the last sub-period), followed by the portfolio containing GI (0.16 over the full rolling sample). GI was the most effective in reducing risk only during the pre-war period. Treasury Bonds and GBs exhibit the least efficacy in risk reduction without substantial differences observed between the two portfolios.

Upon comparing Korean portfolios, the greatest levels of risk reduction were obtained in portfolios containing gold, followed by those containing Treasury Bonds or GBs, across all periods. In the last sub-period, both TBs and GBs, and especially gold, performed better in reducing risk compared to individual portfolios. This was also true for portfolios containing the market index (Table A3, Panel B and Table A4, Panel B). Notably, when employing the optimal weighting strategy, substantial risk reduction was achieved in all portfolios, highlighting the superiority of this approach.

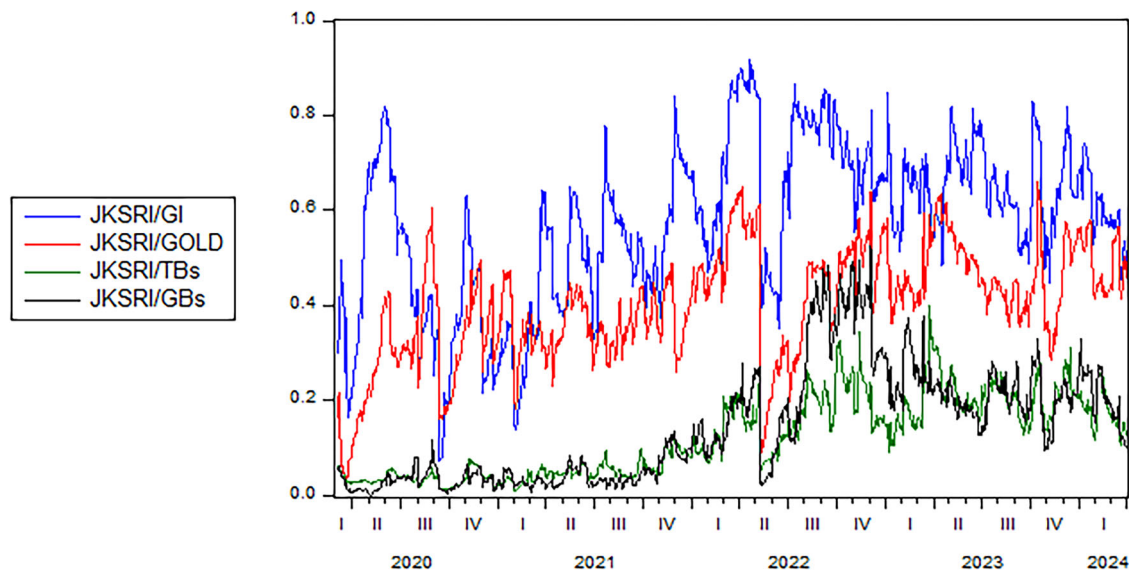


Fig. 6 Rolling one-step-ahead optimal portfolio weights. Indonesia. The optimal weightings refer to GI, Gold, TBs and GBs in the portfolios considered.

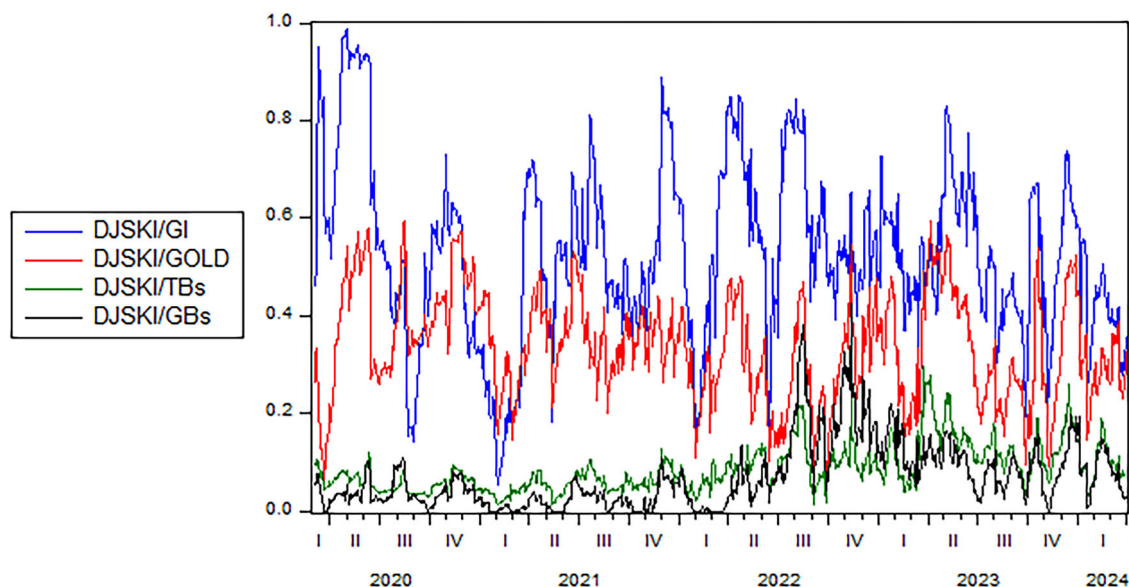


Fig. 7 Rolling one-step-ahead optimal portfolio weights. Korea. The optimal weightings refer to GI, Gold, TBs and GBs in the portfolios considered.

Henceforth, the analysis focuses on examining the diversification benefits from the portfolio that provides the greatest risk reduction, namely the optimally weighed portfolio (Tables 6 and 7 for sustainable indices, and Tables A5 and A6 for market indices). In terms of risk-adjusted returns, only a few optimally weighted portfolios outperform the individual index when considering the entire period. In Indonesia, the commodities portfolio was the most profitable, followed by the gold portfolio. In Korea, the only portfolio that outperforms the individual index in terms of risk-adjusted returns is the one that includes commodities.

However, when the analysis is broken down into sub-periods for sustainable indices, the conclusions differ. During the pre-war period, the inclusion of gold in the portfolio did not improve the risk-adjusted returns of the individual index, despite providing the greatest reduction in risk. Conversely, during the war period, the portfolio containing gold is the most profitable and provides the greatest risk reduction in Indonesia. In Korea, this portfolio results in the smallest losses and the highest risk reduction,

confirming the resilience of portfolios that include gold during periods of uncertainty. These findings are in line with Robiyanto et al. (2023a) and Iglesias-Casal et al. (2020).

It is worth mentioning that in the pre-war period, which coincides with the COVID-19 pandemic, the inclusion of GBs in portfolios with the sustainable index improved the risk-adjusted return of the individual asset. These diversification benefits align with the conclusions of Reboredo (2018) and Reboredo and Ugolini (2020). In Korea, the portfolio that includes treasury bonds also outperforms the individual asset in terms of risk-adjusted returns.

The losses experienced by all portfolios containing the Korean sustainable index during the war period can be partially explained by the depreciation of the South Korean won against the US dollar, which followed the interest rate hikes in the US throughout 2022. Since all variables in our analysis are valued in dollars, this currency effect is relevant. When repeating the analysis for Korea using local currency, the results presented in Table 8 were obtained. In general, the findings for the full rolling

Table 6 Optimal weights portfolio strategy: benefits of diversification.

Portfolio	Mean return	Std.Dev.	Risk-adjusted return	HE
Full rolling sample	(3/03/2020-3/29/2024)			
JKSRI	0.0146	1.5290	0.0095	
JKSRI/GI	0.0369	1.3984	0.0264	0.1635
JKSRI/GOLD	0.0135	1.2462	0.0108	0.3357
JKSRI/TBS	0.0098	1.4293	0.0069	0.1262
JKSRI/GBs	0.0113	1.4441	0.0078	0.1081
Pre-WAR	(3/03/2020-2/23/2022)			
JKSRI	0.0215	1.9838	0.0108	
JKSRI/GI	0.0812	1.6165	0.0502	0.3360
JKSRI/GOLD	0.0064	1.6449	0.0039	0.3124
JKSRI/TBs	0.0197	1.9040	0.0104	0.0789
JKSRI/GBs	0.0214	1.9273	0.0111	0.0562
WAR	(2/24/2022-3/29/2024)			
JKSRI	0.0080	0.9123	0.0088	
JKSRI/GI	-0.0051	1.1551	-0.0059 ^a	-0.6031
JKSRI/GOLD	0.0203	0.6828	0.0297	0.4398
JKSRI/TBs	0.0004	0.7425	0.0006	0.3376
JKSRI/GBs	0.0019	0.7413	0.0025	0.3398

Indonesia Sustainable Index.

The risk-adjusted return is measured by the ratio of the portfolio's mean to its standard deviation for the different portfolios.

^aFor negative returns the risk adjusted return is calculated as the product of portfolio's mean and the standard deviation.**Table 7 Optimal weights portfolio strategy: benefits of diversification.**

Portfolio	Mean return	Std.Dev.	Risk-adjusted return	HE
Full rolling sample	(3/03/2020-3/29/2024)			
DJSKI	0.0186	1.6474	0.0113	
DJSKI/GI	0.0468	1.4298	0.0327	0.2467
DJSKI/GOLD	0.0143	1.2925	0.0111	0.3845
DJSKI/TBs	0.0168	1.5262	0.0110	0.1417
DJSKI/GBs	0.0171	1.5707	0.0109	0.0910
Pre-WAR	(3/03/2020-2/23/2022)			
DJSKI	0.0612	1.8751	0.0326	
DJSKI/GI	0.1084	1.6537	0.0655	0.2223
DJSKI/GOLD	0.0408	1.4791	0.0276	0.3778
DJSKI/TBs	0.0623	1.7675	0.0353	0.1115
DJSKI/GBs	0.0623	1.8330	0.0340	0.0444
WAR	(2/24/2022-3/29/2024)			
DJSKI	-0.0216	1.3992	-0.0302 ^a	
DJSKI/GI	-0.0114	1.1784	-0.0134 ^a	0.2907
DJSKI/GOLD	-0.0106	1.0878	-0.0115 ^a	0.3955
DJSKI/TBs	-0.0262	1.2566	-0.0329 ^a	0.1935
DJSKI/GBs	-0.0257	1.2743	-0.0327 ^a	0.1705

Korea Sustainable Index.

The risk-adjusted return is measured by the ratio of the portfolio's mean to its standard deviation for the different portfolios.

^aFor negative returns the risk adjusted return is calculated as the product of portfolio's mean and the standard deviation.

sample and pre-war period remain consistent, but in the last sub-period, both the portfolio containing GI and the one containing gold outperform the individual asset in terms of risk-adjusted returns, with all returns being positive.

A comparison between portfolios containing the sustainable index and those with the market index shows that, in certain periods, some portfolios with the sustainable index outperform those with the market index. Specifically, this occurs in Indonesia during the last sub-period for the portfolio combined with gold. In Korea, the portfolio that includes GI outperforms the market index portfolio both over the entire period and in the pre-war period. However, this does not apply to the other portfolios, in line with the findings in the literature, which suggest that there are no substantial differences between the performance of sustainable investments and their market benchmarks.

Importantly, it is crucial to clarify that the most profitable portfolios do not always correspond to those with the highest reduction in variance. Furthermore, the combination of gold with either sustainable or market indices proves to be the most advantageous strategy in terms of risk-adjusted returns during the war period, whereas commodities were more favourable in the earlier periods.

Discussion

The empirical findings provide valuable insights into the potential of ESG indices as tools for diversification and risk management in contrasting economic contexts. The analysis confirms that the inclusion of sustainable indices in portfolios, combined with assets like gold and commodities, offers significant diversification benefits. This aligns with the study's objective to assess how these indices perform in mitigating financial risks during periods of

Table 8 Optimal weights portfolio strategy: benefits of diversification.

Portfolio	Mean return	Std.Dev.	Risk-adjusted return	HE
Full rolling sample (3/03/2020-3/29/2024)				
DJSKI	0.0302	1.4206	0.0212	
DJSKI/GI	0.0555	1.3063	0.0425	0.1545
DJSKI/GOLD	0.0298	0.9988	0.0299	0.5057
DJSKI/TBs	0.0206	1.0948	0.0188	0.4061
DJSKI/GBs	0.0249	1.1784	0.0212	0.3120
Pre-WAR (3/03/2020-2/23/2022)				
DJSKI	0.0616	1.6599	0.0371	
DJSKI/GI	0.1081	1.4629	0.0739	0.2233
DJSKI/GOLD	0.0432	1.1789	0.0367	0.4956
DJSKI/TBs	0.0489	1.3022	0.0376	0.3846
DJSKI/GBs	0.0558	1.4073	0.0396	0.2812
WAR (2/24/2022-3/29/2024)				
DJSKI	0.0004	1.1504	0.0004	
DJSKI/GI	0.0057	1.1377	0.0050	0.0218
DJSKI/GOLD	0.0172	0.7926	0.0217	0.5252
DJSKI/TBs	-0.0062	0.8541	-0.0053 ^a	0.4487
DJSKI/GBs	-0.0042	0.9111	-0.0038 ^a	0.3727

Korea Sustainable Index.

All variables are valued in Wons.

The risk-adjusted return is measured by the ratio of the portfolio's mean to its standard deviation for the different portfolios.

^aFor negative returns the risk adjusted return is calculated as the product of portfolio's mean and the standard deviation.

economic stability and global uncertainty, such as the pre-war and war periods analyzed.

The results demonstrate that the A-DCC model effectively captures the dynamic relationships between sustainable indices and other financial assets. For Indonesia, the JKSRI index shows pronounced sensitivity to adverse market events, reflected in positive asymmetric coefficient. This behavior is consistent with the characteristics of emerging markets, which often exhibit higher volatility in response to negative shocks. For Korea, the DJSKI index reveals similar patterns, albeit with reduced volatility, likely attributable to its more advanced regulatory environment and stable market conditions.

The optimal portfolio analysis highlights the need for active management in sustainable investments. Gold contributes the most to improving the risk-adjusted performance of the individual portfolio, particularly during the war period, reinforcing its role as a safe haven asset. Commodities, while effective during the pre-war period, show reduced effectiveness in mitigating risks during heightened uncertainty. These findings align with prior research, such as that by Mensi et al. (2017), which emphasizes the context-dependent nature of asset performance in ESG portfolios.

Green bonds also exhibit noteworthy behaviour, enhancing portfolio performance in the pre-war period across both countries. Their low correlation with other assets aligns with findings by Reboredo (2018), suggesting their utility in multi-asset strategies aimed at balancing risk and return. However, during the war period, their relative effectiveness diminishes, underscoring the necessity of context-specific evaluations for portfolio construction.

An important observation is the variability in hedge ratios and portfolio weights over time, highlighting the dynamic nature of risk and the importance of adaptive strategies. While the portfolios with sustainable indices generally perform on par with those containing market indices, combinations involving gold or commodities frequently outperform during specific periods. This variability underscores the necessity for tailored strategies that account for geopolitical and economic shifts, particularly in markets like Indonesia and Korea.

Overall, the empirical results validate the importance of sustainable indices in constructing resilient and diversified portfolios, demonstrating their utility in addressing the challenges posed by

volatile market conditions. These findings provide actionable insights for investors seeking to integrate ESG principles into their strategies, ensuring alignment with both financial and sustainability objectives.

Conclusions

This paper utilizes an A-DCC model along with out-of-sample analysis to explore dynamic conditional correlations, optimal weights and hedge ratios for sustainable indices for Korea (DJSKI) and Indonesia (JKSRI) when combined with four different assets: commodities (GI), gold and treasury bonds (TBs) futures, and green bonds (GBs). The study uses daily closing prices from 14th October 2014 to 29th March 2024. Using fixed rolling window estimates, we generate 1,064 one-step-ahead forecasts for the period ranging from 3rd March 2020 to 29th March 2024. This allows us to distinguish between the pre-war scenario and the Ukraine war period itself. These results are then compared to those obtained by replacing sustainable indices with market indices.

The findings highlight the leverage effect and volatility clustering in sustainable and market indices across both countries, with long-term persistence in variance and correlation dominating short-term effects. Notably, adverse shocks had a greater impact on volatility than positive ones, confirming asymmetric behavior. Hedging strategies revealed that commodities and green bonds were the most effective instruments for mitigating risk, especially during turbulent periods such as the Ukraine war. However, optimal portfolio weighting consistently outperformed hedging strategies, offering greater risk reduction and stability.

In terms of risk-adjusted returns, portfolios combining sustainable indices with commodities delivered the highest returns over the entire sample period in both countries. During the Ukraine war period, gold emerged as the most resilient asset, particularly in Indonesia, where its inclusion significantly enhanced portfolio performance. Green bonds provided diversification benefits during the pre-war period, especially in Korea, where treasury bonds also contributed to improved risk-adjusted returns. Across all periods, optimal portfolios demonstrated higher variance reduction compared to hedging strategies, emphasizing the need for active portfolio management.

These results underscore the need for tailored policies to promote sustainable finance in diverse economic contexts. Policymakers should focus on increasing the liquidity and accessibility of green bonds, particularly in emerging markets like Indonesia, to facilitate their integration into sustainable investment strategies. The findings also highlight the importance of encouraging alternative assets such as gold and commodities in ESG portfolios, as these assets significantly enhance risk-adjusted performance during global economic disruptions.

Data availability

The data presented in this study are available on request from the corresponding author. The data used in this study were obtained under a specific license from the Bloomberg and Eikon databases, which restricts their public availability. However, they can be requested from the study authors through a reasonable request and with permission from the Bloomberg and Eikon databases.

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Notes

- 1 Beyond financial returns, ESG investing integrates broader goals, such as environmental sustainability, social justice, and ethical governance (Qoyum et al., 2021; Chatzitheodorou et al., 2019; Statman, 2006).
- 2 The results of the diagnostic tests applied to the standardized residuals of the 4 estimated models indicate that in most cases, both autocorrelation and autoregressive conditional heteroscedasticity have disappeared, indicating an adequate model specification. These results are available upon request.

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AIC and CLA: conceptualization, formal analysis, methodology, supervision, data curation, software, validation, writing-reviewing and editing. MCLP and JMMS: conceptualization, formal analysis, visualization, supervision, validation, writing-reviewing and editing. All authors agreed with the final version of the manuscript.

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The authors declare no competing interests.

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Additional information

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